

AHAR, Iran, Islamic Republic of

WMO: 407040

Lat: 38.433N

Lon: 47.067E

Elev: 1390

StdP: 85.70

Time Zone: 3.50 (IRN)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.2	-9.4	-17.6	0.9	-8.9	-15.3	1.2	-6.6	12.2	3.9	11.2	4.3	2.0	290	0.517

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.5	32.5	16.7	30.9	16.5	29.4	16.1	18.7	28.3	18.0	27.3	17.3	26.4	4.9	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.4	13.0	22.5	14.6	12.3	22.1	13.9	11.8	21.6	59.0	28.2	56.5	27.3	54.4	26.3	23.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 10.9	9.5	8.2	DB	-16.3	36.1	3.7	1.5	-19.0	37.2	-21.1	38.1	-23.2	39.0	-25.9	40.1	(4)
(5)			WB	-16.9	19.7	3.3	0.7	-19.3	20.2	-21.3	20.6	-23.1	21.0	-25.5	21.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.6	-0.1	1.6	5.9	10.5	15.5	20.2	22.4	22.5	18.4	12.8	6.6	2.1		
	DBStd	8.84	4.78	4.56	4.95	4.29	2.87	2.50	2.60	2.52	3.22	4.08	4.29	4.83		
	HDD10.0	1116	312	237	142	43	1	0	0	1	21	114	247			
	HDD18.3	2838	570	469	384	236	94	11	3	37	176	351	503			
	CDD10.0	1697	0	0	16	59	172	306	385	386	252	107	13	2		
	CDD18.3	378	0	0	0	2	8	66	129	131	39	3	0	0		
	CDH23.3	3726	0	0	1	29	109	667	1156	1324	398	42	0	0		
	CDH26.7	1204	0	0	0	6	13	190	398	493	103	2	0	0		
(14) Wind	WSAvg	3.5	4.4	4.1	4.1	3.6	3.1	3.2	3.3	3.2	2.8	2.8	3.3	4.0		
(15) Precipitation	PrecAvg	310	22	22	36	48	52	22	10	7	10	26	30	24		
	PrecMax	413	39	55	66	86	99	60	38	45	37	86	94	60		
	PrecMin	223	3	5	17	15	16	1	0	0	0	6	2	6		
	PrecStd	50	9	11	11	20	23	16	10	10	10	19	19	14		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.0	13.4	21.7	27.1	28.3	33.4	35.4	34.6	31.9	26.7	19.3	15.8		
		MCWB	4.7	5.8	9.3	12.6	13.9	15.6	17.1	16.4	15.1	12.6	8.9	6.6		
	2%	DB	9.1	11.5	18.4	22.7	26.2	30.6	32.3	32.6	29.5	24.5	17.2	12.4		
		MCWB	3.4	4.6	8.1	10.9	13.4	16.1	17.6	16.5	14.5	12.1	7.7	5.3		
	5%	DB	7.6	9.7	16.0	20.3	24.3	28.8	30.5	31.0	27.6	22.3	15.2	10.2		
		MCWB	2.6	3.7	7.0	10.2	13.0	15.6	17.2	16.5	14.7	11.5	7.4	4.3		
	10%	DB	6.0	7.9	13.8	18.1	22.5	27.1	28.7	29.4	25.7	20.0	13.3	8.5		
		MCWB	1.6	2.9	6.1	9.4	12.6	15.2	16.7	16.2	14.7	10.7	6.9	3.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.6	6.7	10.0	13.4	16.2	18.5	19.9	19.5	18.4	14.5	10.5	7.4		
		MCDB	9.6	12.4	18.8	23.1	24.4	28.9	30.4	29.1	25.9	22.1	16.1	13.6		
	2%	WB	4.0	5.2	8.7	12.1	15.0	17.4	18.9	18.6	17.1	13.2	9.2	6.0		
		MCDB	8.0	10.0	17.1	20.2	22.9	27.1	29.2	27.7	24.9	20.7	14.4	11.0		
	5%	WB	2.8	4.2	7.5	11.0	14.2	16.6	18.0	18.0	16.3	12.4	8.2	4.9		
		MCDB	6.9	8.7	14.9	18.2	21.5	25.8	27.9	27.1	24.1	19.8	13.4	9.2		
	10%	WB	1.9	3.2	6.5	10.1	13.3	15.9	17.3	17.4	15.6	11.6	7.2	3.8		
		MCDB	5.6	7.5	13.1	16.6	20.3	24.8	26.6	26.4	23.0	18.3	12.3	8.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.5	8.2	9.6	10.6	11.3	12.3	11.3	12.5	11.8	10.5	9.1	7.9		
		MCDBR	8.1	10.3	13.0	14.6	14.4	15.3	14.1	15.5	16.0	14.6	12.5	9.9		
	5% WB	MCWBR	4.7	5.6	5.9	5.9	4.9	5.2	5.0	5.6	5.8	5.7	6.2	5.6		
		MCWBR	7.4	9.3	11.5	12.5	12.6	13.1	12.1	12.9	12.7	12.5	9.7	8.6		
(40) Clear-Sky Solar Irradiance	taub		0.309	0.346	0.403	0.477	0.470	0.454	0.495	0.460	0.409	0.394	0.332	0.305		
	taud		2.331	2.214	2.069	1.905	1.974	2.055	1.963	2.053	2.170	2.155	2.312	2.357		
	Ebn at Noon		862	868	848	805	819	830	792	811	832	800	823	838		
	Edh at Noon		91	118	151	189	180	166	181	162	135	124	93	83		
(44) All-Sky Solar Radiation	RadAvg		2.35	3.21	3.94	4.89	6.18	7.51	7.33	6.68	5.38	3.72	2.49	2.04		
	RadStd		0.22	0.41	0.40	0.51	0.33	0.41	0.36	0.34	0.33	0.33	0.31	0.25		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	N/A	N/A	N/A	+1.08	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page